Using Debugging Tools (e.g., gdb)

Debugging tools are software programs that allow developers to control the execution of a program in a more granular way, inspect memory and variables, and pinpoint exactly where a bug occurs. One of the most powerful debugging tools for C/C++ developers is **gdb (GNU Debugger)**. Let's explore how to use gdb for debugging programs.

Getting Started with gdb

Compiling Code for Debugging To use gdb, you must compile your code with debugging information. In C or C++, you can do this by using the `-g` flag when compiling:

```
gcc -g my_program.c -o my_program
```

1. This flag includes debugging information in the executable, which gdb can then use to give you more details about the program's execution.

Starting gdb To begin debugging with gdb, run the following command in the terminal:

```
gdb ./my_program
```

2. This launches gdb and loads your program for debugging.